A DATABASE MANAGEMENT SYSTEM PROJECT

AI SMART RESUME ANALYZER

Intelligent Analysis. Better Insights. Smarter Hiring

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1. Introduction

Recruiters receive hundreds of resumes for every job opening. Manually reviewing each resume is time-consuming and inefficient.

The **AI Smart Resume Analyzer** is a database system designed to store candidate information, resumes, technical skills, GitHub profiles, and AI-generated analysis results in an organized and structure

This project demonstrates the practical implementation of **Database Management System (DBMS)** concepts using MySQL.

2.Objectives

- Design a normalized relational database.
- Store candidate details securely.
- Manage uploaded resumes.
- Store technical skills.
- Store GitHub profile information.
- Compare resumes with job descriptions.
- Store AI-generated analysis results.

3.System Overview

Modules

- User Management
- Resume Management
- Skills Management
- GitHub Management
- Job Description Management
- AI Analysis Results

Workflow:

```
User
   ↓
Upload Resume
   ↓
Database
   ↓
AI Analysis

   ↓
Analysis Result
```

4.Database Design

Database Name:**AI_Smart_Resume_Analyzer**

Tables:

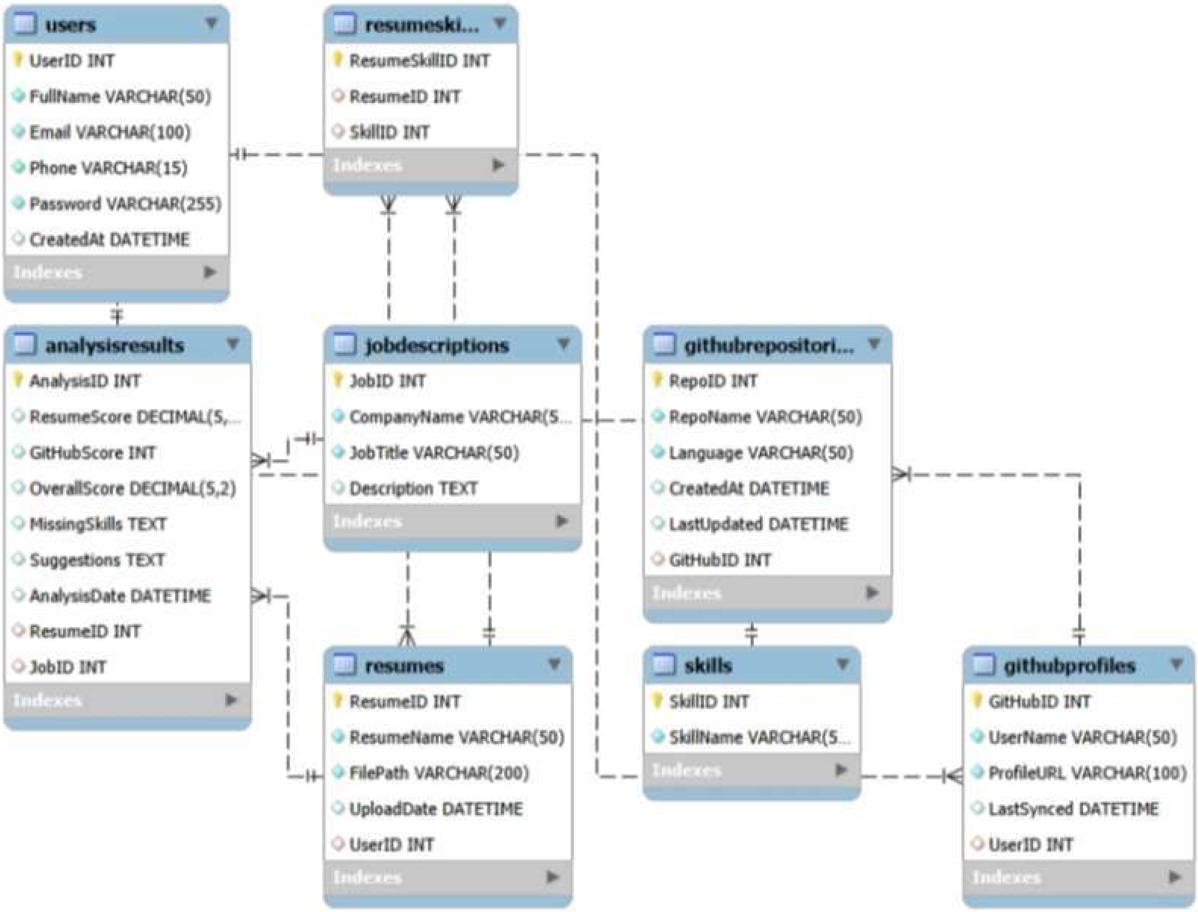
- Users
- Resumes
- Skills
- ResumeSkills
- GitHubProfiles
- GitHubRepositories
- JobDescriptions
- AnalysisResults

Mention:

- Primary Keys
- Foreign Keys
- Relationships

5.ER Diagram

Title: Entity Relationship Diagram



Explain:

- One User → Many Resumes
- One Resume → Many Skills
- One GitHub Profile → Many Repositories
- One Resume → Many Analysis Results

6.SQL Operations Used

The following SQL operations were implemented to develop and manage the **AI Smart Resume Analyzer** database.

- **CREATE DATABASE:** Used to create the `AI_Smart_Resume_Analyzer` database.
- **USE:** Selects the project database for executing SQL commands.
- **CREATE TABLE:** Creates the required tables such as Users, Resumes, Skills, GitHubProfiles, and AnalysisResults.
- **ALTER TABLE:** Modifies the structure of existing tables whenever required.
- **INSERT INTO:** Adds sample records into the database tables.
- **SELECT:** Retrieves data from one or more tables.
- **UPDATE:** Modifies existing records in the database.
- **DELETE:** Removes unwanted records from the tables.

7.Tools and Technologies

The **AI Smart Resume Analyzer** was developed using the following tools and technologies:

- **MySQL:** Used as the relational database management system to store and manage project data.
- **MySQL Workbench:** Used for database design, SQL query execution, and ER diagram creation.
- **SQL:** Used to create database objects and perform data manipulation and retrieval operations.
- **Git:** Used for version control and tracking project changes.

8.Advantages

- Automates the storage and management of candidate resume data.
- Organizes user, resume, skills, GitHub, and job information in a structured database.
- Reduces manual effort in managing and analyzing resumes.
- Enables efficient resume-to-job matching using relational data.
- Stores AI-generated resume scores, missing skills, and improvement suggestions.

9.Future Scope

The AI Smart Resume Analyzer can be enhanced with additional features to improve its functionality and user experience. Some possible future improvements are:

- Integrate AI and NLP for automatic resume parsing.
- Implement an AI-based resume ranking system.
- Provide personalized job recommendations based on candidate skills.
- Develop a web application with a user-friendly interface.
- Integrate cloud databases for secure and scalable data storage.
- Add real-time GitHub profile synchronization using APIs.

11. Conclusion

- Successfully designed and implemented the **AI Smart Resume Analyzer** database using **MySQL**.
- Applied core **DBMS concepts** such as normalization, primary keys, foreign keys, and table relationships.
- Managed resume, skills, GitHub, job description, and analysis data in a structured manner.
- Performed data management using essential **SQL operations**.
- Created a scalable database that can be integrated with an AI-powered resume analysis application in the future.
- Strengthened practical database design and SQL skills through this portfolio project.